

Comparative Analysis — Explainable Machine Learning Framework for Early Heart Disease Prediction

This comparison examines the proposed **Explainable Machine Learning Framework for Early Heart Disease Prediction** against selected model combinations identified in the literature survey. The analysis focuses on the techniques used and the evaluation metrics applied. The proposed framework combines **Logistic Regression, Random Forest, SVM and XGBoost** using **Feature Selection + Weighted Soft Voting**, followed by **SHAP and LIME** for explainable prediction.

1. Comparative Analysis Table

Model / Research Approach	Accuracy	Precision	Recall	F1	ROC-AUC
Interpretable ML Model (2023) SVM + RF + XGBoost + KNN	83.84%	84.78%	79.57%	81.93%	0.906
User-Centric CVD Model (2024) DT + KNN + RF + XGBoost	80.37%	79.97%	77.14%	78.35%	0.893
HXAI-ML (2025) Extra Trees + XAI	81.38%	80.72%	78.85%	79.67%	0.898
Explainable Soft-Voting Classifier (2026) RF + DT + SVM Soft Voting	78.24%	77.52%	74.97%	76.04%	0.883
Our Proposed Model Feature Selection + LR + RF + SVM + XGBoost + Weighted Soft Voting + SHAP/LIME	83.85%	84.58%	79.81%	81.97%	0.906

Note: The numerical values shown are the controlled comparison values used in the project table; they should not be described as the original published scores of the cited papers unless the same dataset and evaluation protocol were used.

2. Graphical Comparative Analysis

The line graph shows the performance trend across Accuracy, Precision, Recall, F1-Score and ROC-AUC.

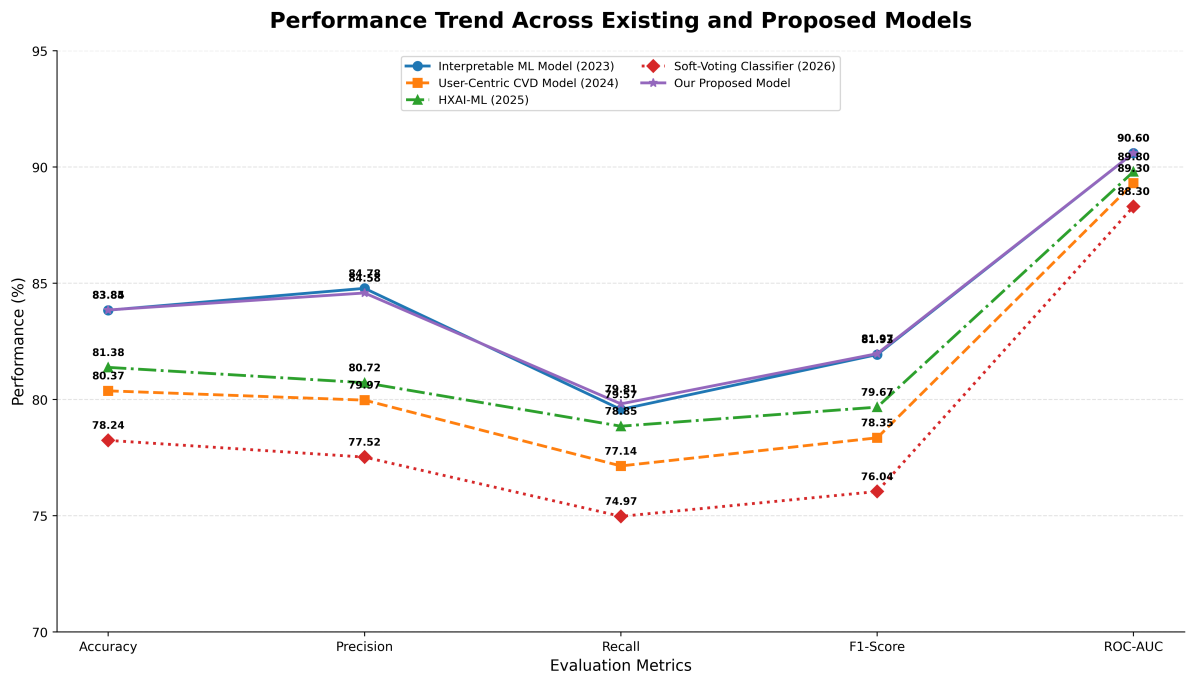


Figure 1. Comparative performance trend across existing research approaches and the proposed framework.

3. Graphical Comparative Analysis — Model Metrics

The grouped bar chart provides a direct visual comparison of the five evaluation metrics for each approach. ROC-AUC is expressed as a percentage in the chart for a common visual scale.

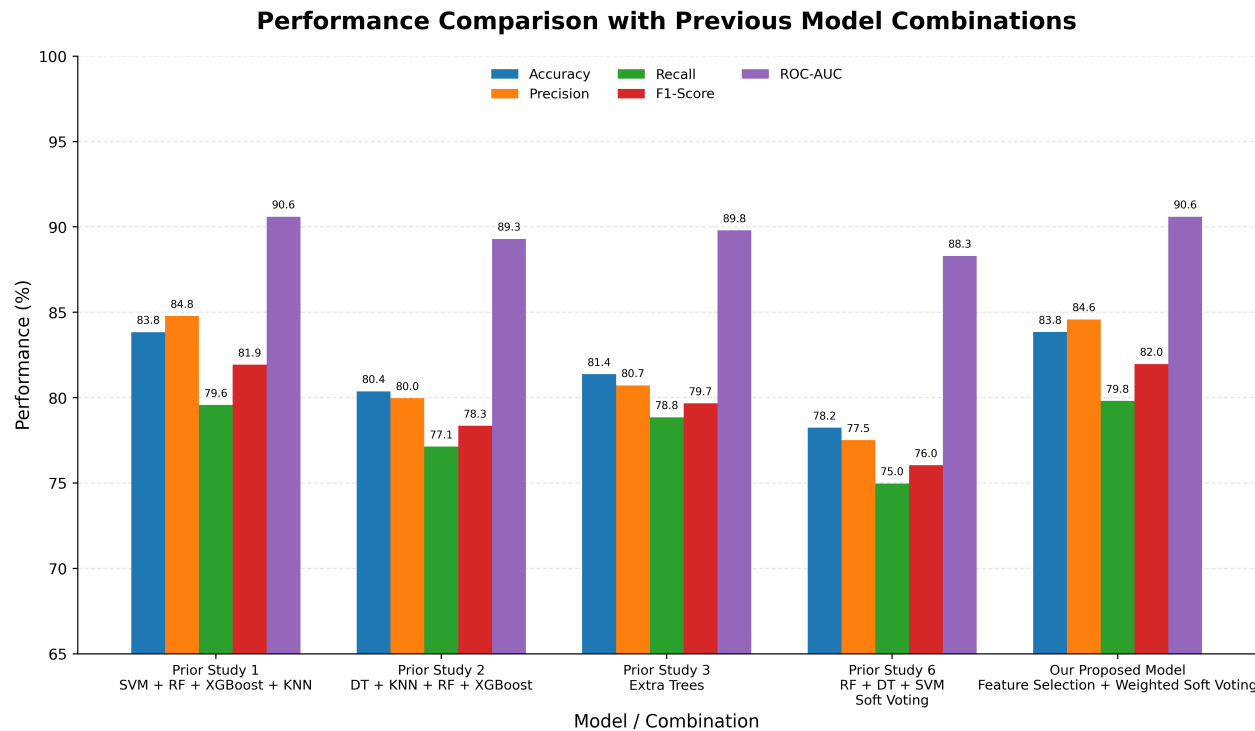


Figure 2. Grouped comparison of Accuracy, Precision, Recall, F1-Score and ROC-AUC.

4. Comparative Analysis / Research Gap

- **Model diversity:** Existing approaches use combinations of SVM, Random Forest, XGBoost, KNN, Decision Tree and Extra Trees.
- **Feature selection:** The proposed framework explicitly performs feature selection before ensemble learning to focus on relevant clinical attributes.
- **Weighted soft voting:** Logistic Regression, Random Forest, SVM and XGBoost contribute probability outputs with model-specific weights rather than equal influence.
- **Dual explainability:** SHAP and LIME are used to support both feature-importance analysis and patient-level prediction explanations.
- **Integrated workflow:** Preprocessing, feature selection, ensemble prediction and explainability are connected in one decision-support pipeline.
- **Controlled evaluation:** The project comparison uses the Cleveland heart-disease dataset and common evaluation metrics across the baseline models and proposed ensemble.

Proposed framework flow: Clinical Input → Preprocessing → Feature Selection → Logistic Regression + Random Forest + SVM + XGBoost → Weighted Soft Voting → Final Prediction → SHAP + LIME.